

Measuring and Picture Graphs

Answer Key

Kindergarten–Grade 1

Quick Answers

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|---------------|---------------|-------|
| 1. 6 | 5. 3 | 9. — |
| 2. — | 6. 18 | 10. — |
| 3. <i>Ana</i> | 7. <i>Leo</i> | |
| 4. 3 | 8. 4, 6, 9 | |
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Chart data used throughout: Mia 5, Leo 3, Ana 6, Sam 4 stickers. **Measuring strip:** crayon 4, marker 6, pencil 7, straw 9 paper clips.

1. How many stickers does Ana have?

Find Ana's row, then touch and count each star once: 1, 2, 3, 4, 5, 6. Ana's row has six stars, and the key says one star means one sticker, so the count *is* the number of stickers: 6.

Listen for: a child who counts a star twice, or who reads the row above.

2. Write $>$ or $<$: pencil ___ crayon.

The strip says the pencil is 7 paper clips and the crayon is 4 paper clips. Both were measured with the *same* paper clips, so the bigger count is the longer object. Seven is more than four, and the symbol opens toward the bigger side: $>$.

Say it out loud: “pencil is greater than crayon.”

3. Who has the most stickers?

Compare the rows by length, not by name: Mia 5, Leo 3, Ana 6, Sam 4. The longest row of stars is Ana's, with 6 — more than 5, 4, and 3. So Ana has the most.

4. How many more stickers does Ana have than Leo?

“How many more” means compare the two rows and count the extra. Line them up: Ana 6, Leo 3. Three of Ana's stars match Leo's three; the leftover stars are the extra. $6 - 3 = 3$, so Ana has 3 more stickers.

Listen for: answering “6” (the whole row) instead of the difference.

5. How many paper clips longer is the straw than the marker?

Same “how many more” idea, now with lengths. Straw is 9 paper clips, marker is 6 paper clips, and both used the same paper clip. Count on from 6 to 9: 7, 8, 9 — that is three counts. $9 - 6 = 3$, so the straw is 3 paper clips longer.

6. How many stickers do all four children have in all?

“In all” means put every row together. Add the four counts, making tens where you can: $5 + 3 = 8$, then $8 + 6 = 14$, then $14 + 4 = 18$. Altogether they have 18 stickers.

7. Who has the fewest stickers?

“Fewest” asks for the *shortest* row, which is the opposite of problem 3. Mia 5, Leo 3, Ana 6, Sam 4: the shortest row is Leo’s, with only 3 stars. So Leo has the fewest.

8. Order crayon, marker, and straw from shortest to longest.

Read each length off the strip: crayon 4, marker 6, straw 9 paper clips. Because the same unit was used every time, ordering the objects is just ordering the numbers, smallest first: 4, then 6, then 9. So the order is crayon (4), marker (6), straw (9).

9. Is Mia right that Leo + Sam beats Ana + Mia?

Do not judge by the claim — get both totals from the chart first. Leo and Sam: $3 + 4 = 7$ stickers. Ana and Mia: $6 + 5 = 11$ stickers. Now compare the two totals: 7 is less than 11, so Mia’s claim is wrong and the correct symbol is $<$.

Listen for: a child who compares single children (Sam 4 vs Mia 5) instead of the two totals.

10. Same straw: 9 paper clips but only 5 blocks — which unit is longer? (open response — read the reasoning)

The straw never changed length; only the measuring unit changed. It took *fewer* blocks to cover the same straw, so each block must cover more of it than a paper clip does. Fewer units to cover one length always means the units are bigger. Therefore one block is longer.

A full-credit answer includes a picture or a sentence such as: “The straw is the same both times. It only took 5 blocks but 9 paper clips, so one block is longer than one paper clip.” A drawing showing 5 long blocks and 9 short clips under the same straw earns the same credit.

Not full credit: “9 is bigger than 5, so paper clips are longer” — that compares the counts instead of the units.